

Anish Vempaty

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Summary

Software engineer with experience building automation frameworks and CI/CD pipelines for test execution using Python and Linux environments. Developed backend utilities and scripts to automate diagnostics, manage resources, and streamline integration across cloud and emulation platforms. Skilled in Git workflows, containerization, and test management, seeking to apply automation expertise to infrastructure roles.

Education

New York University

MS, Cybersecurity (GPA: 3.7/4.0)

- **Achievements:** NYU Scholarship, Gyandhan Scholarship

Aug 2023 - May 2025

Brooklyn, NY

SRM Institute of Science and Technology (SRMIST)

B.Tech, Computer Science (GPA: 9.27/10)

Jun 2019 - May 2023

Chennai, India

Skills

- **Security and Forensics:** DFIR, Threat Hunting, SIEM (Splunk), Pen Testing, Incident Response, MITRE ATT&CK, IDS, Wireshark, Snort, Burp Suite
- **Networking:** Routing and Switching (Cisco), VLANs, OSPF, ACLs, VPN, Network Security, Firewalls, Packet Analysis
- **Cloud and DevOps:** AWS (Lambda, S3, SQS, SES, DynamoDB, API Gateway), Docker, Kubernetes, GitHub Actions, CI/CD, AWS CDK, Job Scheduling Systems - SLURM
- **Programming and ML:** Python, C/C++, C#, Java, SQL/NoSQL, TensorFlow, PyTorch, Scikit-learn, OpenCV, REST APIs, FastAPI, Flask, .net, Bash, Node.js, React
- **AI and GenAI:** LangChain, HuggingFace, LLMs, RAG, MLflow, GenAI Security Tools
- **Other:** Linux (ARM/x86), Spark, Automation, System Design, Data Pipelines (Kafka, Spark), Log Analysis
- **Test Automation and Verification:** Test Automation Frameworks, Hardware Verification, Chip Development Workflows, Emulation Environments - ZeBu, Emulation Environments - QEMU, Regression Testing

Experience

GreyOrange / ROBO

Robotics Security & Operations Engineer

Feb 2026 - Present

Atlanta, GA

- Conduct network penetration testing and implement regression testing automation frameworks using Bash and Python to secure robot-to-cloud communication and prevent unauthorized access to AMR fleets.
- Vulnerability Management: Review and debug robot code in Python, Java, and C++ to identify security flaws and ensure robust, bug-free software deployment.
- AI-Driven Operations: Utilize AI-assisted analytics and telemetry logs to detect system anomalies and forecast operational risks, ensuring high reliability at the SAMs ATL site.
- Develop backend utilities and scripts using Bash to automate diagnostics, integrate robotic systems with Warehouse Management Systems, and build CI/CD pipelines for continuous deployment.

Delta Rising Foundation

Data Scientist & Team Co-Leader

Aug 2025 - Feb 2026

Remote

- Co-led a 6-member team to quantify per-plot Giant Sequoia carbon sequestration, engineering proprietary datasets with Google Earth Engine and multi-source satellite imagery, which provided the foundation for accurate carbon estimates across the study area
- Developed robust ML models using TensorFlow and Scikit-learn and built an interactive C# frontend that let users visualize predictions and explore carbon estimation tools, improving user access to the analysis

SRMIST

Data Engineer Intern

Jun 2022 - Dec 2022

Chennai, India

- Automated alumni data collection using Python/Selenium orchestrated through SLURM job scheduling, processing 30+ years of records.
- Designed and implemented ETL pipelines in Python (Pandas, SQL) deployed via AWS CDK with CI/CD, increasing dataset accuracy and reliability for analytics.

Foxmula Corp

Cybersecurity Intern

Dec 2020 - Feb 2021

Bengaluru, India

- Developed AES + chaos-based image encryption reducing runtime by 10%.
- Led a 4-member team improving throughput and performance by 5%.

Verzeo Tech

AI Engineer Intern

Apr 2020 - Jun 2020

Bengaluru, India

- Built a YOLOv3-based accident detection app improving accuracy from 84% to 93%.

- Designed a Flask dashboard with async logging to ensure high system uptime.

Projects

Edge Device Cyber Threat Detection Using ML

- Built a real-time intrusion detection system on Linux-based Raspberry Pi using Python, Scapy, and implemented SVM and Random Forest models
- Automated deployment and monitoring with Bash scripts and Docker, and visualized live network anomalies via Streamlit dashboard for on-device monitoring

Cloud-Based Source Code Vulnerability Detector

- Developed a serverless GitHub vulnerability scanner using AWS Lambda, Python, and integrated async SQS workflows and DynamoDB caching
- Configured CI/CD pipeline with Git, GitHub Actions, and Bash scripts to enable automated vulnerability scanning and AI-driven remediation via Gemini API

Custom Cybersecurity LLM

- Fine-tuned a Qwen2.5 7B model on curated cybersecurity datasets for threat and secure code analysis.
- Integrated an optimized GPU-accelerated inference pipeline with quantized deployment.

RISC-V C910 Microarchitecture Redesign for x86 Compatibility

Present

- Redesigned an open-source RISC-V C910 core to introduce microarchitectural changes that improve execution compatibility and performance for x86-targeted software workloads.
- Modified hardware interfaces and instruction-handling logic to reduce x86 dependency by enabling more efficient execution and translation of x86-oriented binaries on RISC-V hardware.
- Implemented and verified ALU and instruction decode (IDU) modifications in Verilog using Icarus Verilog (iverilog) and QEMU-based co-simulation, improving effective x86 workload execution speed on the RISC-V C910 core by ~40%.